

the drains/gates of the NMOS transistors when clock goes high. Neither PMOS device will turn on until either M1 or M2's drain moves below $V_{DD} - V_{THP}$. For low power (low contention current), try to keep one side of the latch shut off until a significant imbalance is present in the circuit.

Removing Sense Amplifier Memory

Figure 16.30 shows how the sense amplifier's memory can be erased. M1–M4 form a latch (Fig. 13.16). To remove the sense amplifier's memory, *all* nodes in the sense amplifier must be actively driven to a known voltage (no floating or dynamically charged nodes). When *clock* is low, the sense amplifier's outputs are pulled to V_{DD} through MS3 and MS4. The MOSFETs MS1 and MS2 are off, breaking the connection between V_{DD} and ground (so no current flows in the latch). The gates of M1 and M2 are at V_{DD} so their drains are actively driven to ground. In other words, when *clock* is low, all nodes in the circuit are either pulled high to V_{DD} or low to ground.

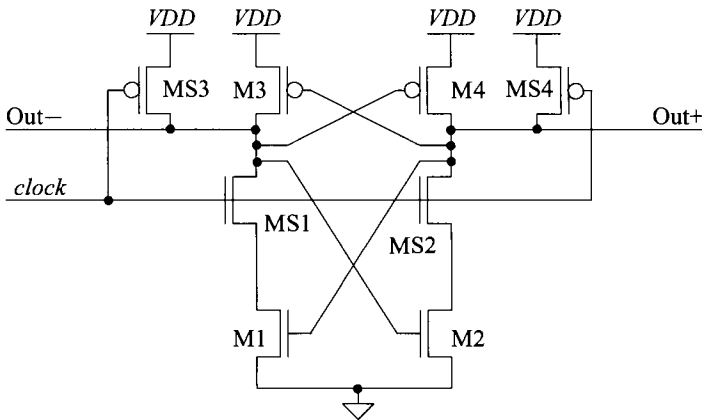


Figure 16.30 Removing memory in a sense amplifier.

Creating an Imbalance and Reducing Kickback Noise

When *clock* goes high, the latch action can take place. For the comparison to function as desired, we need to generate an imbalance. This, alone, isn't too difficult. What is difficult is creating an imbalance for a wide range of input signals while not causing an excessive amount of current to flow in the circuit. Consider the additions to our sensing circuit seen in Fig. 16.31. MB1/MB2 can be connected to either the drain or the source of MS1/MS2, as seen in the figure. Let's first consider connecting them to the drains. When *clock* is low, the drains of MB1 and MB2 are pulled to V_{DD} . If the input voltages are significant, a *large* current can be drawn from V_{DD} (when *clock* is low). If the input voltages are relatively small, then the current pulled from V_{DD} may be tolerable (keeping in mind that the sense amplifier doesn't function correctly when the input voltages are less than V_{THN}). The benefit of connecting to the drains is high gain. Very small voltage differences can cause quick sensing (the outputs swing quickly to valid logic voltages). Note that the possibility for large current flowing through either MB1/MB2 is present independent of the state of *clock*.

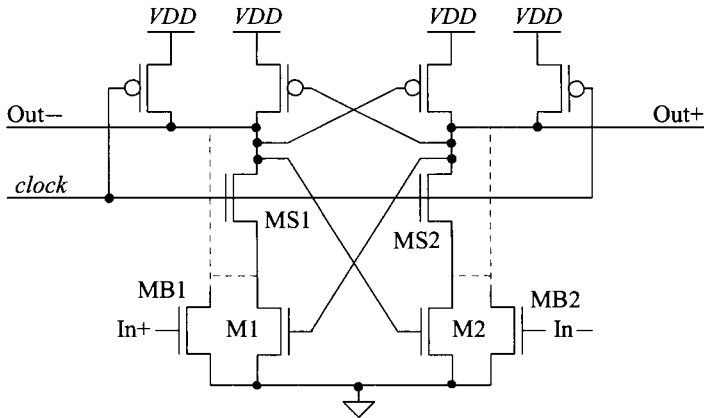


Figure 16.31 Creating an imbalance in a sense amplifier.

Next let's consider connecting MB1/MB2 to the sources of MS1/MS2 (drains of M1/M2). When *clock* goes low, the drains of MB1/MB2 are pulled low through M1/M2. No current flows in the comparator when *clock* is low (unlike when the drains of MB1/MB2 are connected to the drains of MS1/MS2). The gain is much lower because now MB1/MB2 are operating in the triode region when *clock* goes high. The potential for large current flow is still present when *clock* is high but, now, the maximum value on the drain of MB1/MB2 is $V_{DD} - V_{THN}$. And so the potential for MB1/MB2 to move into the triode region can result in a smaller output current.

We seem to have a problem. To reduce the possibility of metastability, we need to increase the gain in series with the signal path (increase the W/L ratio of MB1 and MB2; that is, reduce their switching resistance). However, this results in larger power dissipation. How can we keep power down while maximizing the sensitivity of our sense amplifier? While we can use long L devices for MB1 and MB2 so they never pull significant current (at the cost of sensitivity and speed) or have a separate pre-amplifier to generate the input signals, let's look at some other ideas (noting that no design is perfect but a trade-off between power, speed, and sensitivity).

Figure 16.32 shows one idea for creating an imbalance without the possibility of significant current flow. MB1 and MB2 are used to create an imbalance in the gate-source voltages of M1/M2. Since, prior to switching, the voltage on the gates of M1/M2 is V_{DD} , we can still have good sensitivity even though MB1 and MB2 are operating in the triode region (look like resistors). The large voltage dropped across $V_{GS,M1}$ and $V_{DS,MB1}$ ensure that good gain. It is important, however, that the inputs signals are $> V_{THN}$ to ensure that the circuit operates properly. The sensing fails if this isn't the case because neither MB1 or MB2 can turn on and provide the needed path to ground for proper logic level signals. As long as the inputs are above the threshold voltage, there aren't any dynamic nodes (*no memory in the sense amplifier*). The sources of MS1 and MS2 are still pulled to ground when *clock* is low. Kickback noise is reduced using this topology because the inputs are isolated from the latch by MB1/MB2. Kickback noise is still present and must be considered when designing and simulating this sense amplifier.

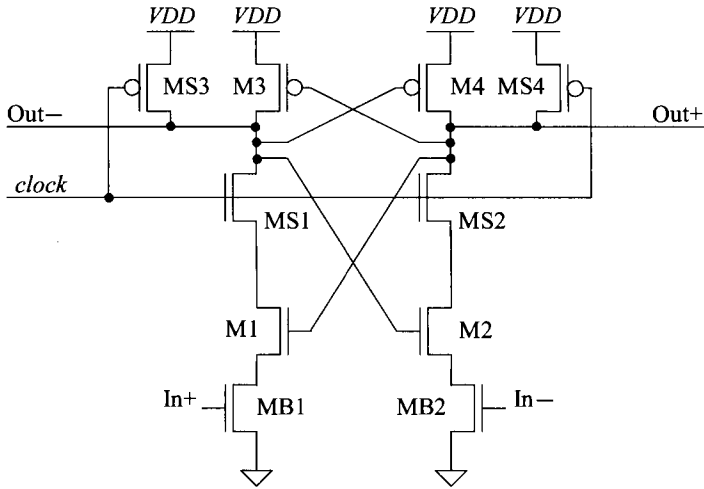


Figure 16.32 Reducing power in a sense amplifier.

Note that we might try to design our comparator to operate over a wide range of input voltages by adding PMOS devices in series with the sources of M3/M4, as seen in Fig. 16.33. We know that MB1–MB4 must be capable of turning on when *clock* goes high. If either pair can't turn on, then the latch won't be able to generate full logic levels. For the design in Fig. 16.33, the input signals would have to fall within V_{THN} and $VDD - V_{THP}$ (more restricted than the circuit in Fig. 16.32). Also, and perhaps more importantly, MB3/MB4 can't generate an imbalance in the latch. If the sources of M1/M2 are connected to ground so that MB1/MB2 don't generate the imbalance, then when *clock* goes high, MB3 and MB4 are off. They won't turn on until the NMOS pair M1/M2 turns on and drops the gates of M3/M4 below $VDD - V_{THP}$. For great differences in the input signals, the comparator may still function correctly. However, for the majority of the input signal differences, the resulting output logic levels is determined by the matching between M1/M2, e.g., the one with the lower "actual" V_{THN} will turn on faster.

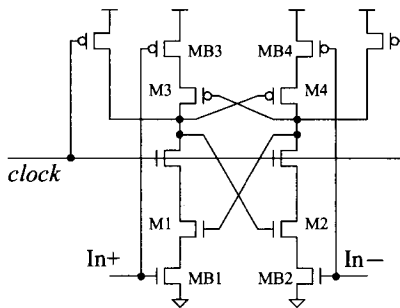


Figure 16.33 A bad design: how not to get wide-swing operation.

Example 16.1

Suppose that it is suggested that the gates of M3/M4 be tied to the drains of M1/M2 or MB1/MB2, respectively, in the sense amplifier (clocked comparator) seen in Fig. 16.32. Comment on the concerns.

If the gates of M3/M4 are tied to the drains of M2/M1, then, when *clock* goes high, a significant contention current flows in the latch. As discussed earlier, it is desirable that one side of the latch (either the NMOS or PMOS pairs) is off when the latch is clocked in order to reduce power. The same concerns are present when these gates are connected to the drains of MB1/MB2.

We might try to eliminate MS3/MS4 when we tie the gates of M3/M4 to the drains of MB1/MB2. Because the drains of MB1/MB2 always move to ground when *clock* is low (again assuming the inputs are $> V_{THN}$), M3/M4 will turn on and pull the outputs high. When *clock* goes high, though, it's impossible to shut M3/M4 off. Because the maximum voltage on the sources of MS1/MS2 is $V_{DD} - V_{THN}$, the gates of M3/M4 can't go to V_{DD} to shut one of the devices off. ■

Increasing the Input Range

Figure 16.34 shows adding, to our basic sense amplifier in Fig. 16.32, MOSFETs MB3–MB8 so that the circuit will function with input signals ranging from 0 to V_{DD} (actually it will operate correctly with input signals beyond the power supply rails). We know current can only flow out of the sources of M1/M2, so our additional circuitry, for creating the imbalance, must be capable of sinking current from these sources (and so we'll add MB3 and MB4 for this reason). Next, we know that MB1/MB2 function fine for creating the imbalance if the input signals are above V_{THN} . However, if the input signals are below V_{THN} , they turn off. To level-shift the input, let's use the MB5–MB8. If the input signals are $< V_{THN}$, then MB7 and MB8 are on. A difference in the input voltages causes different currents to flow in MB5 and MB6. The different currents flowing in these transistors results in different voltages across each one. This voltage difference is then used to create an imbalance in MB3 and MB4. Unfortunately, the addition of MB5–MB6 contradicts our earlier statement that, “*For minimum power it's important that there are no DC paths from V_{DD} to ground except during switching times.*” There is a DC path through MB8/MB6 and MB7/MB5. To lower the power, we can increase the length of MB5 and MB6 so that the continuous current flowing through the DC path is lessened.

Simulation Examples

Let's simulate the operation of the circuit in Fig. 16.32. We know that the outputs of the sense-amp go high every time *clock* goes low. To make the outputs change only on the rising edge of *clock*, we can use the SR latch discussed in Ch. 13 on the output of the circuit, Fig. 16.35. Now, with the addition of the NAND gates, when the sense amplifier's outputs are high, the outputs of the latch don't change from the previous decision (made on the rising edge of *clock*).

Figure 16.36 shows the current supplied by V_{DD} for the circuit in Fig. 16.35. While the current seen in this figure includes the current supplied to the NAND gates, it still should be compared to Fig. 16.29. Notice in Fig. 16.36 that current flows only during the times the clock changes states (unlike the current flow in Fig. 16.29).

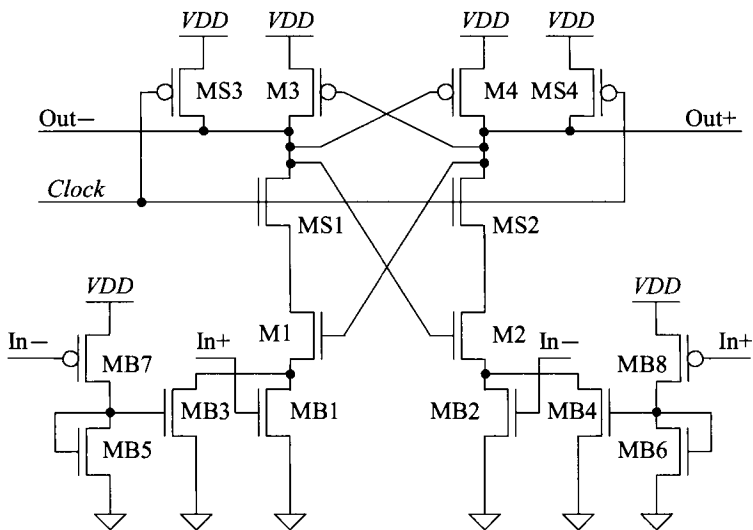


Figure 16.34 Rail-to-rail input signal range. See also Fig. 27.16.

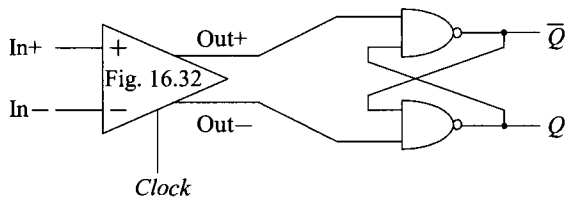


Figure 16.35 Adding an SR latch so output only changes on rising edge of the clock.

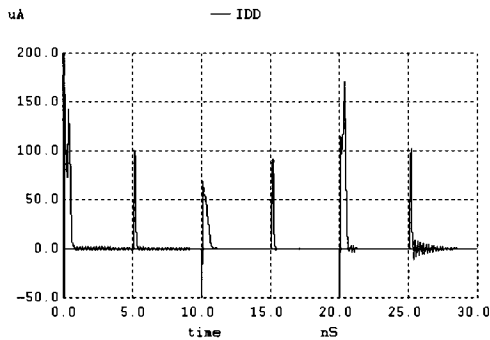


Figure 16.36 Regenerating the plot seen in Fig. 16.29 with the sense amplifier from Fig. 16.35. The current includes the switching current from the NAND gates.

Next let's look at the kickback noise. The clock feedthrough noise is small for this topology because the *clock* is isolated from the input by three transistors. Let's put the sense amplifier in the topology seen in Fig. 16.28 where the inputs are driven to the rails by long-length inverters. The simulation results are seen in Fig. 16.37. The kickback noise is approximately 5 mV.

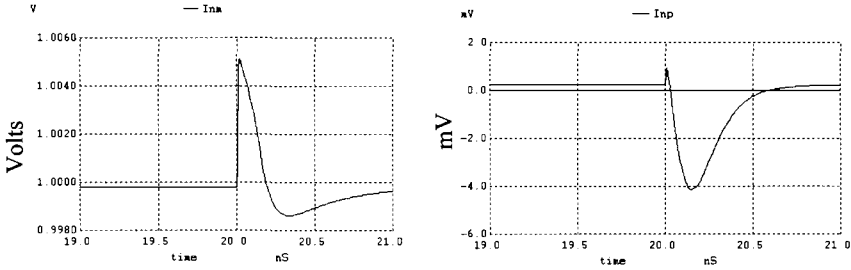


Figure 16.37 Kickback noise.

Figure 16.38 shows the operation of the circuit in Fig. 16.35 where the outputs only change on the rising edge of *clock*. The input signals are the same ones that were used in generating Fig. 16.27 (although the simulation was 20 ns longer in Fig. 16.38). The sensitivity of this sense amplifier is better than 10 mV. The sensitivity can be improved by increasing the lengths of MB1/MB2 (so that they have a large resistance). The drawback of the improved sensitivity is the increased time it takes to pull one of the outputs to ground (the output transition time). The sensitivity can be increased without this penalty by adding a preamplifier (e.g., a differential amplifier) between the sense amplifier and the inputs, see Fig. 27.16.

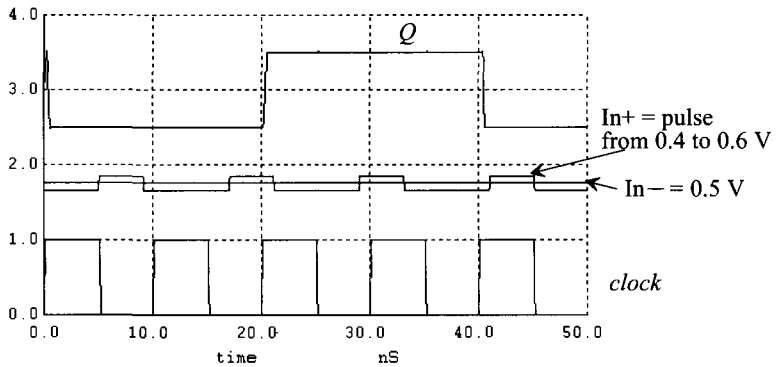


Figure 16.38 The operation of the circuit in Fig. 16.35.

16.2.2 Row/Column Decoders

Reviewing the memory block organization in Fig. 16.25, we may wonder how to address one or more of the 256kbit subarrays. When we say “address,” for the row lines, what we mean is that we are driving one, and only one, of the row lines in one or more of the 256k subarrays to a voltage greater than $V_{DD} + V_{THN}$ (to fully turn the access MOSFET on), while holding all other row (word) lines at ground. To accomplish this selection, a *row address decoder* is used. Once the word line has transitioned high and the sensing is complete, the data from the memory cells is present on the bit lines (as discussed in detail in Sec. 16.1). To select the addressed data from one or more of these bit lines (in one or more memory arrays), a *column decoder* is used.

In a large (capacity) memory the chip’s address pins are multiplexed, that is, the same set of pins are used for both the row and column addresses. A separate clock signal is used to strobe in either the row or the column addresses, at different times, into some hold registers. In older DRAMs, for example, the falling edge of $\overline{RAS}$ (row address strobe) was used to clock in the row address and the falling edge of $\overline{CAS}$ was used to clock in the column address. The outputs of the row address hold register (column address hold register) are decoded and used to select a row (column) line, Fig. 16.39.

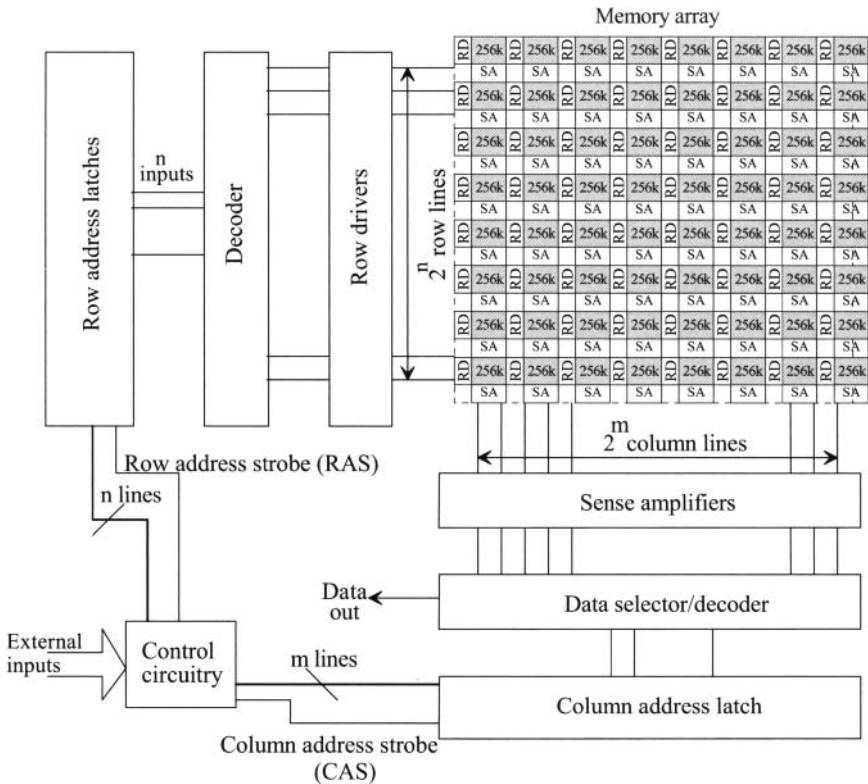


Figure 16.39 Detailed block diagram of a RAM.

The length of the column (m-bits in the Fig. 16.39) and row (n-bits in Fig. 16.39) address words depends on the size of the memory and the word size. For example, a 1-Gbit memory can be organized in x1 (by one), x2 (by two), x4 (by four), etc., configuration. The “by four” for example, simply signifies the word size used in the memory’s input/output data path is 4-bits. A 1-Gbit memory in a x4 configuration simply indicates that 256-Mwords can be addressed where each word is 4-bits. In a x1 configuration 1-Gbits of data can be accessed ($2^{30} = 1\text{-G} = 1,073,741,824$). We would then have 15 address pins for 2^{15} different row and column addresses (remembering that the address pins are shared). The next question then becomes how to decode the addresses and select the appropriate row(s) and column(s).

Global and Local Decoders

Looking at Fig. 16.39, we see that we decode the outputs of the row address latches and the column address latches and then feed the outputs to the subarrays. This is called a *global decode*. If the memory size is, once again, 1-Gbit (in a x1 configuration), then 2^{15} ($= 32,768$) row line wires and 2^{15} column line wires are fed to the memory arrays. If each memory array is 256-kbits (512 word lines and 512 column lines), then we need 4,096 arrays to get a 1-Gbit memory (64 by 64 memory arrays in Fig. 16.39). When the output of the row decoder goes high, it turns on a word line in the 64 memory arrays of the addressed row. A total of 32k columns then have data sitting on them. Because our memory is going to take only 1 bit of data and feed it to the output, we have, perhaps, wasted a lot of power. (In many topologies this large amount of data is called a “page.” Once the row is opened it can be quickly read out of the memory by simply changing the column address.) The other issue with the global decoder is that a separate layer of metal is needed to route the decoded signals throughout the entire chip (adding process complexity). The big benefit of global decoding is reduced chip size.

A *local decoder* takes the input addresses and, as the name says, decodes them locally at the memory array. The 15 bits, for the example given above, are routed to each subarray so that they can be decoded locally. This takes up a considerable amount of space on the chip (having a decoder at each 256k array, for example) but doesn’t call for increased process complexity. In most practical designs a combination of local and global techniques are used. Some of the address bits are decoded globally while others are decoded locally. The global decoder enables an array or (arrays), while the local decoder selects the array’s row line and column line. A static decoder is seen in Fig. 16.40. Figure 16.41 shows how the 10-bit address (for 1,024 word or row lines) is connected to each decoder element.

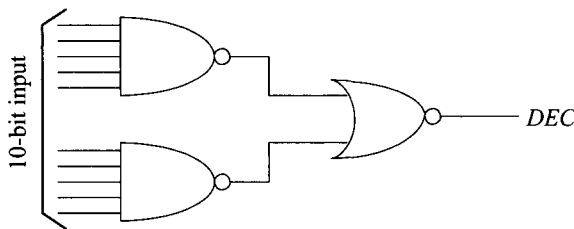


Figure 16.40 A static decoder.

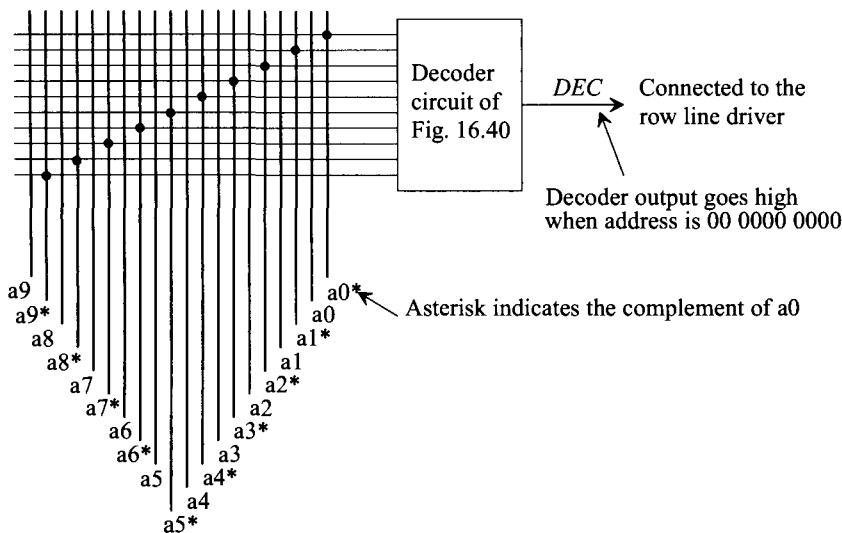


Figure 16.41 How the address lines are connected to a decoder element.

For the column decoder, we know that we have to be able to write or read data to the memory cells (not just select a column). For this reason pass transistors are used on the output of the decoder, as seen in Fig. 16.42. The pass transistor doesn't pass a logic one to full levels. We lose a V_{THN} with body effect when writing or reading a one. The sense amplifier may be used to pull the bit line up to a full logic one. Similarly, on the output of the column decoder a sense amplifier may be used to pull the output line to full logic levels. Finally, note that we drew the column decoder and its outputs going horizontally. In practice, the outputs of the column decoder run in the same direction as the columns themselves in order to save space.

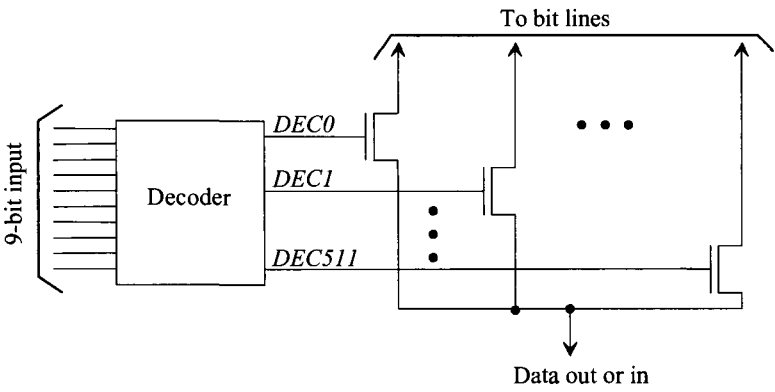


Figure 16.42 Addition of pass transistors to implement a column decoder.

Reducing Decoder Layout Area

To reduce the layout area of address decoders, a pass-transistor based decoder, see Fig. 16.43a, can be used. The concerns with using this decoder are that the unselected outputs are not actively driven high or low (they are floating) and, once again, as NMOS devices they don't pass a logic one to a full logic level (that is, V_{DD}). This means that on the output of the decoder, as part of our row driver, we need a circuit that, when not selected, pulls the output of the decoder low. At the same time, to maximize the noise margins, the switching point of the driver needs to be reduced. For example, with $V_{DD} = 1\text{ V}$ and $V_{THN} = 0.35\text{ V}$ (high because of body effect), the selected output of the decoder swings from 0 to 0.65 V. The switching point of the row driver should lie in the middle of this swing at 0.3125 (ideally), to maximize the noise margins, Fig. 16.43b.

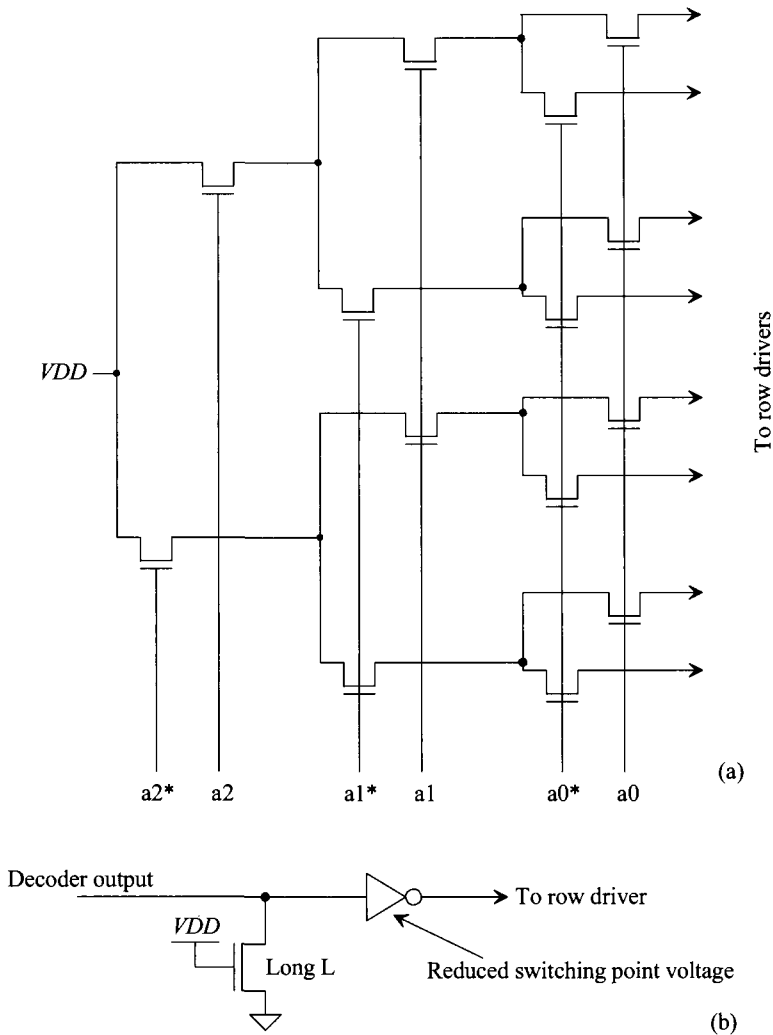


Figure 16.43 A tree decoder used to reduce layout area.

Another technique to reduce decoder layout size uses the precharge-evaluate (PE) logic, Fig. 16.44, discussed in Ch. 14. This figure shows a 3-bit input decoder. The output of the circuit goes high when the input address is 000. Using the long-length keeper MOSFET makes the dynamic circuit operate as a static circuit. Variations of this circuit are common in commercially available memories.

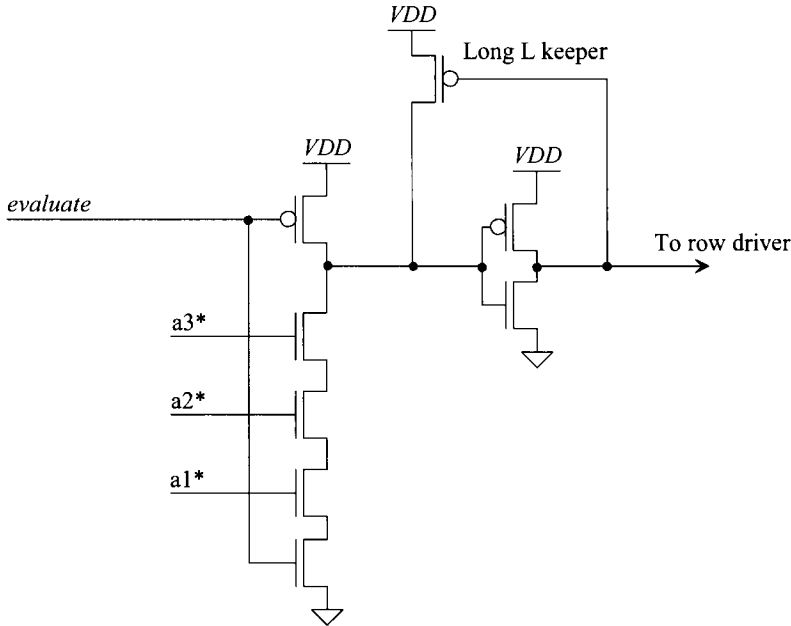


Figure 16.44 3-bit PE decoder.

16.2.3 Row Drivers

In most memories a single NMOS pass transistor switches data into or out of the memory cell. To fully turn this pass transistor on, the row line is driven to a pumped voltage (a voltage outside of the power supply rails that is generated using the on-chip charge pump circuit discussed in Ch. 18). We'll call this pumped voltage $VDDP$. For our purposes this voltage must be greater than $VDD + V_{THN}$ (with body effect). In the following discussion we'll assume a $VDDP$ of 1.5 V (since our VDD is 1 V and V_{THN} is 280 mV without body effect).

Examine the inverter circuit seen in Fig. 16.45. The input to the inverter connected to the word line swings from 0 to VDD (1 V). Note that the PMOS is sitting in its own well. Both its source and body are tied to $VDDP$. When the input to this inverter is low (0 V), the PMOS is on and the NMOS is off. The word line is driven to $VDDP$ (which is what we want). Unfortunately when this inverter input is high ($= VDD$), the NMOS turns on but the PMOS can't shut off. The source gate voltage of the PMOS is 0.5 V (above the PMOS's threshold voltage). In this section we discuss word-line drivers, knowing that a simple inverter can't be used.